SMART LIBRARY REQUEST WORKFLOW IN SERVICENOW:

Requirement Analysis:

1. Customer journey map
2. Data flow diagrams and user stories
3. Solution requirements
4. Technology stack

1.Customer Journey Map: Smart Library Request Workflow:

Field	Details
Date	
Team ID	
Project Name	**Smart Library Request Workflow in ServiceNow**
Maximum Marks	4 Marks

Customer Journey Map: Smart Library Request Workflow:

Phase	Entice	Enter	Engage	Exit	Extend
Goal	Awareness of the Smart Library portal.	Initiating the digital request.	Management and fulfillment of the request.	Completion and item pickup/access.	Feedback and future engagement.
Steps	User hears about the Smart Library via internal comms or the ServiceNow dashboard.	User navigates to the Service Catalog and selects the "Smart Library" category.	User fills out the request form, selects the asset, and submits it for approval.	User receives a notification that the item is ready and collects it from the smart locker.	User returns the item or requests an extension via the portal.
Interactions	Digital: Email announcement, ServiceNow homepage banner.	Digital: ServiceNow Service Portal interface, search bar.	People: Automated approval flow or manual librarian review.	Places: Physical Smart Locker station or Digital Download link.	Things: Mobile app notification, "Return" barcode scanner.
Goals & Motivations	"Help me find the resources I need for my project quickly".	"Help me submit this request without a long manual form".	"Help me track the status of my request in real-time".	"Help me access my requested item securely and at my convenience".	"Help me stay organized with my due dates".
Positive Moments	Easy-to-find link on the main landing page.	Form auto-populates user data (department, ID) from ServiceNow profile.	Instant confirmation email with a reference number.	24/7 access to physical lockers via QR code.	One-click renewal option in the portal.
Negative Moments	Unclear where the "Library" section is within a massive Service Catalog.	Too many mandatory fields or confusing "Required Date" logic.	Waiting for manual approval without a clear ETA.	Locker is malfunctioning or the QR code won't scan.	Receiving late-fee notifications for items already returned.
Areas of Opportunity	Use AI-driven "Recommended for you" based on previous searches.	Integrate "Live Chat" support directly on the request form.	Implement automated approvals for low-cost or digital items.	Add a "Navigate to Locker" map feature within the mobile app.	Create a "Wishlist" feature for items currently checked out.

2.Data Flow Diagram (DFD)and User Stories:

Project Overview:

Field	Information
Date	
Team ID	
Project Name	Smart Library Request Workflow in ServiceNow
Maximum Marks	4 Marks

Data Flow Diagram (DFD): Smart Library Workflow

➔ **A DFD represents how information flows through the ServiceNow system, including data entry, storage, and processing**

High-Level Flow Description:

1. **User** logs into the ServiceNow Portal and selects the Smart Library catalog item.
2. **Request Data** (User ID, Asset ID) is sent to the **Service Catalog Process**.
3. **ServiceNow Database** checks asset availability.
4. **Approval Workflow** triggers an notification to the **Librarian/Manager**.
5. **Fulfillment System** sends a release code to the **User** and updates the **Asset Registry**.

User Stories:

➔ Following the industry-standard template, below are the user stories for the Smart Library workflow:

User Type	Functional Requirement (Epic)	User Story Number	User Story / Task	Acceptance Criteria	Priority	Release
Employee (End User)	Requesting	USN-L1	As a user, I want to search for library assets in the ServiceNow portal so I can find specific resources.	Search results return accurate available items.	High	Sprint-1
Employee (End User)	Requesting	USN-L2	As a user, I want to submit a request for an item so that it is reserved for my use.	User receives a "REQ" number upon submission.	High	Sprint-1
Librarian (Approver)	Approval	USN-L3	As an approver, I want to see the requester's department and project code to validate the request.	Approval record shows all necessary user metadata.	Medium	Sprint-1
System Admin	Automation	USN-L4	As an admin, I want the system to automatically mark an asset as 'Unavailable' once a request is approved.	Database status changes from 'In Stock' to 'Reserved'.	High	Sprint-2
Employee (End User)	Notification	USN-L5	As a user, I want to receive a push notification when my item is ready for pickup.	Notification is sent to the ServiceNow mobile app.		

3.Solution Requirements:

Field	Information
Date	
Team ID	
Topic	Smart Library request workflow in ServiceNow
Maximum Marks	4 Marks

Functional Requirements:

→These requirements define the specific behaviors and features the ServiceNow workflow must perform.

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)
FR-1	User Authentication	Registration through Corporate SSO (Single Sign-On).
		Login via ServiceNow Mobile App.
FR-2	Asset Management	Browse Smart Library Catalog items by category.
		Check real-time inventory availability in the ServiceNow database.
FR-3	Request Workflow	Submission of digital request form for physical/digital assets.
		Automated manager approval routing based on user profile.
FR-4	Fulfillment	Confirmation via ServiceNow notification (Push/Email).
		Smart Locker release code generation upon approval.

Non-functional Requirements:

➔**These requirements define the system's operational standards and quality constraints.**

NFR No.	Non-Functional Requirement	Description
NFR-1	Usability	The interface should follow ServiceNow's Polaris UI standards for intuitive navigation.
NFR-2	Security	Multi-factor authentication (MFA) must be enforced for asset pickup authorization.
NFR-3	Reliability	The request workflow must maintain 99.9% success rate for database status updates.
NFR-4	Performance	Catalog item search results should load in under 2 seconds.
NFR-5	Availability	The Smart Library portal must be accessible 24/7 for global users.
NFR-6	Scalability	The system must handle up to 5,000 concurrent asset requests during peak hours.

4.Technology Stack (Architecture & Stack):

Field	**Information**
Date	**06 April 2026**
Team ID	**SL-REQ-2026**
Project Name	**Smart Library request workflow in ServiceNow**
Maximum Marks	**4 Marks**

Technical Architecture:

➔ **The architecture follows a cloud-based SaaS model using ServiceNow as the core platform, integrating local smart locker hardware.**

Table 1: Technology Stack

S.No	Component	Description	Technology
1.	User Interface	How user interacts with the Smart Library application.	Service Portal (HTML, CSS, AngularJS).
2.	Application Logic-1	Request Fulfillment and Workflow Engine.	ServiceNow Flow Designer (JavaScript).
3.	Application Logic-2	Notification and Messaging Logic.	ServiceNow Notification Engine.
4.	Application Logic-3	Smart Locker integration logic.	REST API Integration Hub.
5.	Database	Core Asset and User Data storage.	ServiceNow MariaDB (SaaS).
6.	Cloud Database	Cloud-based logging and audit trails.	ServiceNow Cloud Data Store.
7.	File Storage	Digital asset and document storage.	ServiceNow Attachment Storage.
8.	External API-1	Identity verification for asset checkout.	Corporate SSO / LDAP API.
9.	External API-2	Notification dispatch to external channels.	Twilio / Slack API.
10.	Machine Learning Model	Smart search and asset recommendation.	ServiceNow Predictive Intelligence.
11.	Infrastructure	Cloud-based SaaS deployment.	ServiceNow Cloud Infrastructure.

Table 2: Application Characteristics :

S.No	Characteristics	Description	Technology
1.	Open-Source Frameworks	Frameworks used for front-end rendering.	AngularJS, Bootstrap.
2.	Security Implementations	Access controls and data encryption standards.	OAuth 2.0, AES-256 Encryption, Role-Based Access Control (RBAC).
3.	Scalable Architecture	Justification of the multi-tier scalability.	3-tier SaaS Architecture (ServiceNow).
4.	Availability	Reliability and server distribution.	High Availability (HA) Pair Nodes, Geo-Redundancy.
5.	Performance	Handling request volume and caching.	ServiceNow Content Delivery Network (CDN), Edge Encryption.

Executive Conclusion: Project Design Phase-II

1.Customer Journey:

➔Mapped the user experience from initial awareness (**Entice**) to long-term engagement (**Extend**).
Identifies pain points and "Areas of Opportunity" to ensure the workflow is not just functional, but delightful for the end-user.

2.Data Flow (DFD) and User Stories:

➔Visualized the movement of data between the User, ServiceNow Process Engine, and physical Smart Lockers.
Ensures data integrity and provides a clear roadmap for backend developers to handle information exchange and storage.
➔Defined specific functional requirements from the perspective of Customers, Librarians, and Administrators.
Establishes clear "Acceptance Criteria" and priorities (High/Medium/Low) to guide Agile development sprints.

4.Solution Requirements:

➔Categorized requirements into Functional (what the system does) and Non-functional (how the system performs).
Guarantees that the solution is not only feature-rich but also secure, scalable, and highly available.

5.Technology Stack:

➔Outlined the 3-tier architecture using ServiceNow’s cloud infrastructure and integration APIs.
Provides the technical specifications (AngularJS, REST APIs, MariaDB) required to build and maintain the system.